

AWS Systems Manager Session Manager vs Bastion Hosts

AWS Systems Manager Session Manager is a capability within AWS Systems Manager that allows you to securely connect to EC2 instances **without using SSH keys, opening port 22, or deploying bastion hosts**.

Instead of routing traffic through a public jump server, Session Manager creates a secure, audited connection through the AWS Systems Manager Agent (SSM Agent) installed on the instance.

How Session Manager Works

1. The EC2 instance runs the **SSM Agent**
2. The instance has an IAM role allowing Systems Manager access
3. You connect through:
 - AWS Console
 - AWS CLI
 - or Session Manager plugin
4. Traffic is tunneled securely through AWS Systems Manager infrastructure

No public IP or bastion host is required.

Comparison Note

Feature	Bastion Host Approach	Session Manager Approach
Public Access	Requires public IP + SSH port	No public IP required
SSH Keys	Requires <code>.pem</code> key management	No SSH keys needed
Port 22	Must be opened	No inbound ports required
Auditing	Limited/manual logging	Built-in session logging
Security	Higher attack surface	More secure by default
Maintenance	Bastion server must be managed	No bastion server needed

3 Advantages of Session Manager Over Bastion Hosts

1. Improved Security

Session Manager eliminates the need to:

- expose port 22
- assign public IPs
- manage SSH keys

This greatly reduces the attack surface.

2. Centralized Auditing & Logging

All sessions can be logged to:

- Amazon CloudWatch
- Amazon S3

This provides better compliance and monitoring compared to traditional SSH access.

3. Lower Operational Overhead

With Session Manager:

- no bastion host to maintain
- no patching jump servers
- no managing inbound SSH rules

This simplifies infrastructure management and reduces cost.

Conclusion

Session Manager is considered the modern best practice for secure EC2 administration because it provides:

- stronger security
- easier management
- centralized auditing
- private access without internet exposure

It is increasingly replacing bastion hosts in modern AWS environments.